

HDOC-8

- Q.1 Write a C program to find the volume, lateral surface area, and total surface area of a triangular pyramid when its base area, base perimeter, slant height, and height are given.

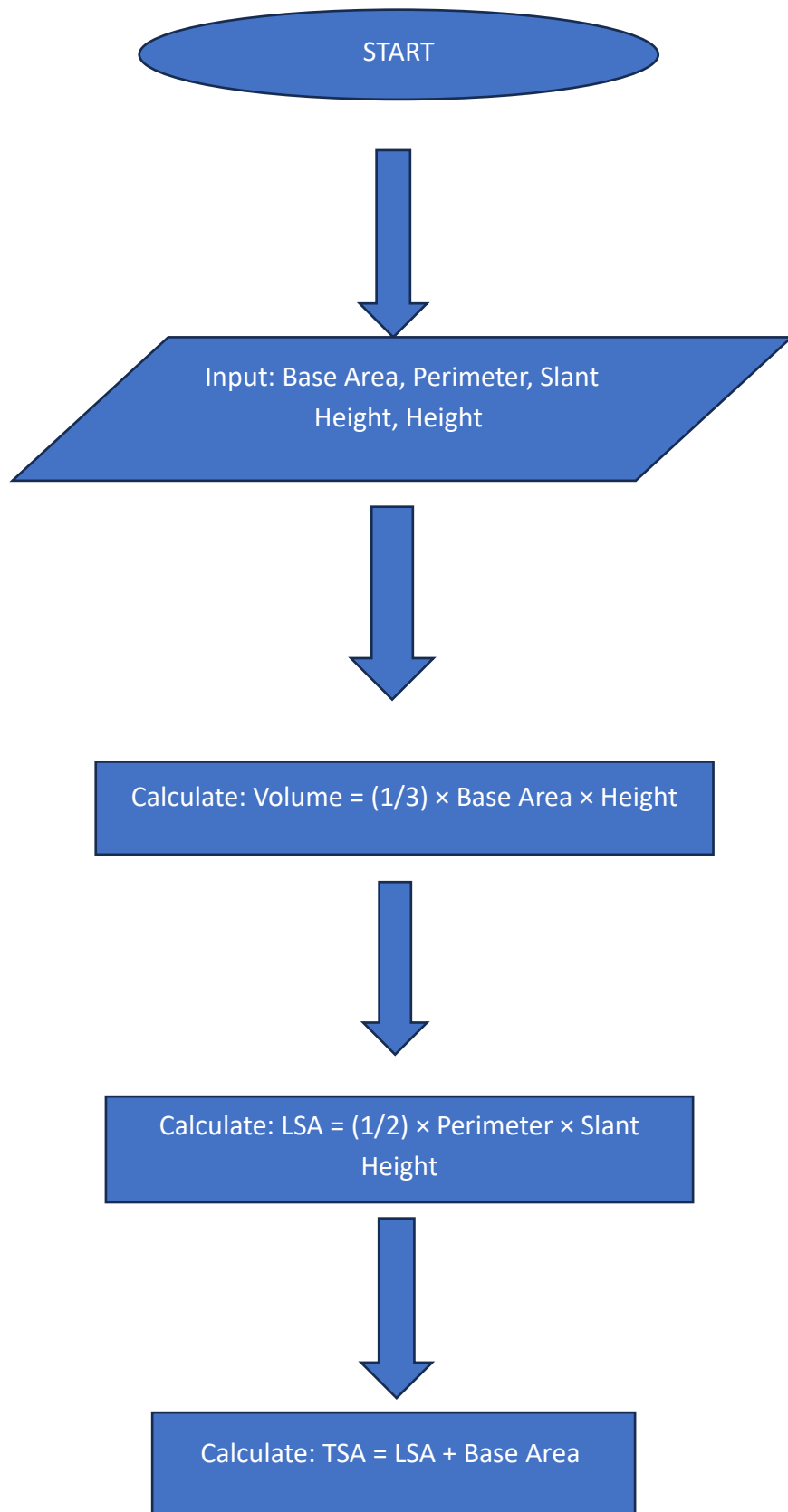
TEST CASE TABLE

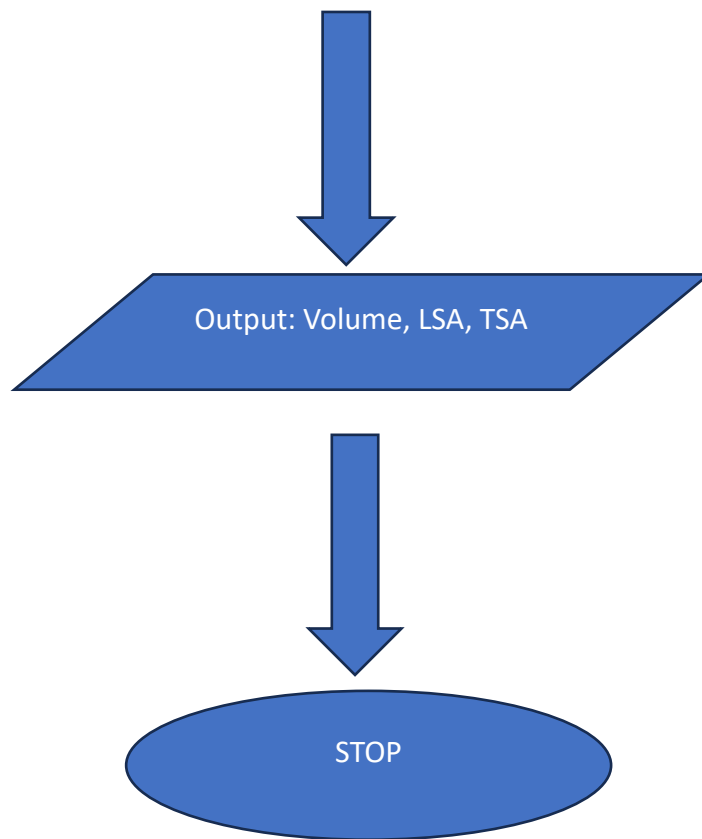
Test case	Inputs(A, P, L, H)	Expected output	Actual output	Result
1	10, 15, 6, 5	V = 16.67, LSA = 45, TSA = 55	V = 16.67, LSA = 45, TSA = 55	PASS
2	20, 24, 8, 10	V = 66.67, LSA = 96, TSA = 116	V = 66.67, LSA = 96, TSA = 116	PASS
3	15, 18, 7, 6	V = 30, LSA = 63, TSA = 78	V = 30, LSA = 63, TSA = 78	PASS

ALGORITHM:

1. Start.
2. Declare float variables: basearea, baseperimeter, slantheight, height, volume, lsa, tsa.
3. Read values for basearea, baseperimeter, slantheight, and height from the user.
4. Calculate $\text{volume} = (1.0 / 3.0) * \text{basearea} * \text{height}$.
5. Calculate $\text{lsa} = (1.0 / 2.0) * \text{baseperimeter} * \text{slantheight}$.
6. Calculate $\text{tsa} = \text{lsa} + \text{basearea}$.
7. Print volume, lsa, and tsa rounded to 2 decimal places.
8. Stop.

FLOWCHART:





C- SOURCE CODE

```
#include <stdio.h>
```

```
int main() {
```

```
    float base_area, base_perimeter, slant_height, height;
```

```
    float volume, lsa, tsa;
```

```
    // Step 1: Input
```

```
    printf("Enter Base Area: ");
```

```
    scanf("%f", &basearea);
```

```
    printf("Enter Base Perimeter: ");
```

```
    scanf("%f", &baseperimeter);
```

```
    printf("Enter Slant Height: ");
```

```
    scanf("%f", &slantheight);
```

```
    printf("Enter Height: ");
```

```
scanf("%f", &height);  
volume = (1.0 / 3.0) * basearea * height;  
lsa = (1.0 / 2.0) * baseperimeter * slantheight;  
tsa = lsa + basearea;  
printf("\n--- Results ---\n");  
printf("Volume: %.2f\n", volume);  
printf("Lateral Surface Area (LSA): %.2f\n", lsa);  
printf("Total Surface Area (TSA): %.2f\n", tsa);  
return 0;  
}
```

Compilation and execution

```
(kali㉿kali)-[~]
$ gedit hdoc8.1.c

** (gedit:11297): WARNING **: 10:41:25.805: Cou
** (gedit:11297): WARNING **: 10:41:25.805: Cou

(kali㉿kali)-[~]
$ cc -g hdoc8.1.c

(kali㉿kali)-[~]
$ ./a.out
Enter Base Area: 10
Enter Base Perimeter: 15
Enter Slant Height: 6
Enter Height: 5

— Results —
Volume: 16.67
Lateral Surface Area (LSA): 45.00
Total Surface Area (TSA): 55.00

(kali㉿kali)-[~]
$ ./a.out
Enter Base Area: 20
Enter Base Perimeter: 24
Enter Slant Height: 8
Enter Height: 10

— Results —
Volume: 66.67
Lateral Surface Area (LSA): 96.00
Total Surface Area (TSA): 116.00

(kali㉿kali)-[~]
$ ./a.out
Enter Base Area: 15
Enter Base Perimeter: 18
Enter Slant Height: 7
Enter Height: 6

— Results —
Volume: 30.00
Lateral Surface Area (LSA): 63.00
Total Surface Area (TSA): 78.00

(kali㉿kali)-[~]
$ █
```

Q.2 Write a C program to find each angle when two complementary angles are in a given ratio.

TEST CASE TABLE

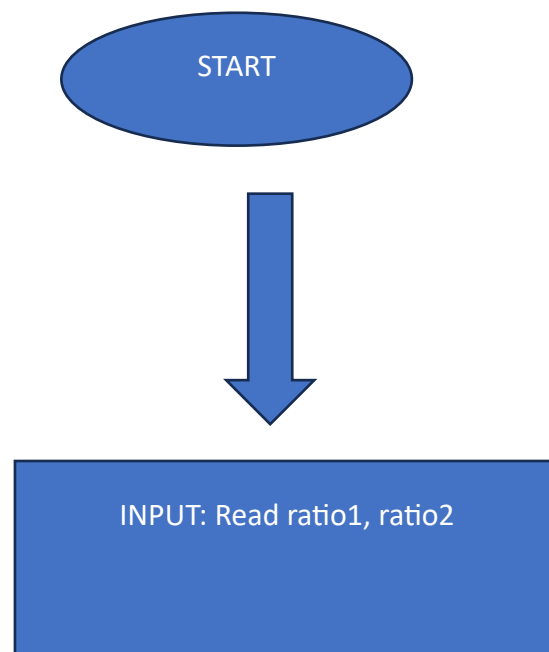
TEST CASE	INPUTS	Expected output	Actual output	Result

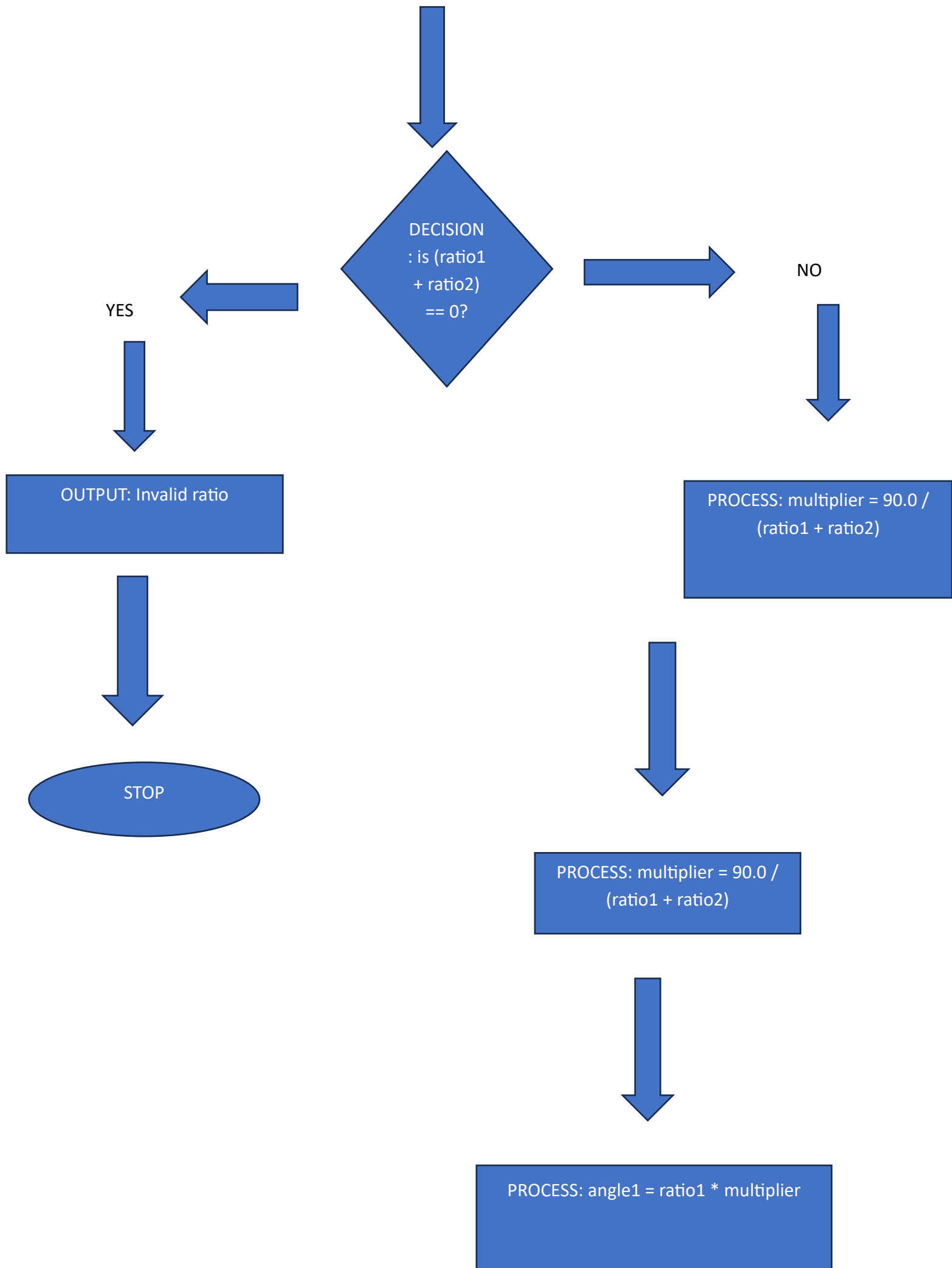
1	1, 1	Angle 1 = 45 Angle 2 = 45	Angle 1 = 45 Angle 2 = 45	pass
2	2, 3	Angle1 = 36 Angle2 = 54	Angle1 = 36 Angle2 = 54	pass
3	4, 5	Angle 1 = 40 Angle 2 = 50	Angle 1 = 40 Angle 2 = 50	pass

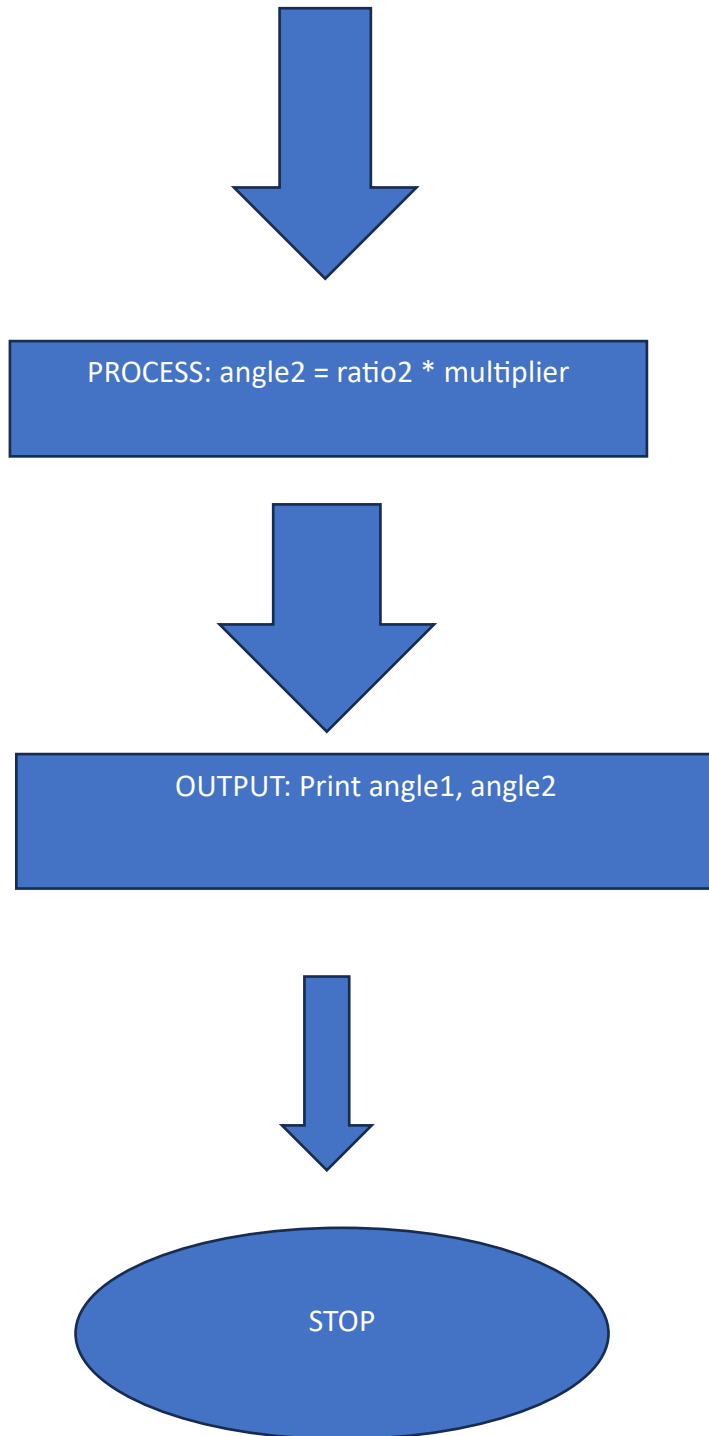
ALGORITHM:

1. Start.
2. Declare float variables: ratio1, ratio2, angle1, angle2, multiplier.
3. Read ratio1 and ratio2 from the user.
4. Check if $\text{ratio1} + \text{ratio2} == 0$. If true, print an error and Stop.
5. Calculate $\text{multiplier} = 90.0 / (\text{ratio1} + \text{ratio2})$.
6. Calculate $\text{angle1} = \text{ratio1} * \text{multiplier}$.
7. Calculate $\text{angle2} = \text{ratio2} * \text{multiplier}$.
8. Print angle1 and angle2.
9. Stop.

FLOWCHART:







C- PROG SOURCE CODE

```
#include <stdio.h>
```

```
int main() {
```

```
    float ratio1, ratio2;
```

```
    float angle1, angle2, multiplier;
```

```
    printf("Enter the first part of the ratio: ");
```



```
scanf("%f", &ratio1);
printf("Enter the second part of the ratio: ");
scanf("%f", &ratio2);
if ((ratio1 + ratio2) == 0) {
    printf("Error: Sum of ratios cannot be zero.\n");
    return 1;
}

multiplier = 90.0 / (ratio1 + ratio2);
angle1 = ratio1 * multiplier;
angle2 = ratio2 * multiplier;
printf("\n--- Results ---\n");
printf("Angle 1: %.2f degrees\n", angle1);
printf("Angle 2: %.2f degrees\n", angle2);

return 0;
}
```

COMPILATION / EXECUTION

```
1 #include <stdio.h>
2
3 int main() {
4     float ratio1, ratio2;
5     float angle1, angle2, multiplier;
6
7     printf("Enter the first part of the ratio: ");
8     scanf("%f", &ratio1);
9     printf("Enter the second part of the ratio: ");
10    scanf("%f", &ratio2);
11
12    if ((ratio1 + ratio2) == 0) {
13        printf("Error: Sum of ratios cannot be zero.\n");
14        return 1;
15    }
16
17    multiplier = 90.0 / (ratio1 + ratio2);
18    angle1 = ratio1 * multiplier;
19    angle2 = ratio2 * multiplier;
20
21    printf("\n— Results —\n");
22    printf("Angle 1: %.2f degrees\n", angle1);
23    printf("Angle 2: %.2f degrees\n", angle2);
24
25    return 0;
26 }
27
28
```

```
(kali㉿kali)-[~]
$ gedit hdoc8.2.c

** (gedit:27693): WARNING **: 12:59:34.942: Could not
** (gedit:27693): WARNING **: 12:59:34.943: Could not

(kali㉿kali)-[~]
$ cc -g hdoc8.2.c

(kali㉿kali)-[~]
$ ./a.out
Enter the first part of the ratio: 1
Enter the second part of the ratio: 1
Trash
— Results —
Angle 1: 45.00 degrees
Angle 2: 45.00 degrees

(kali㉿kali)-[~]
$ ./a.out
Enter the first part of the ratio: 2
Enter the second part of the ratio: 3
— Results —
Angle 1: 36.00 degrees
Angle 2: 54.00 degrees

(kali㉿kali)-[~]
$ ./a.out
Enter the first part of the ratio: 4
Enter the second part of the ratio: 5
— Results —
Angle 1: 40.00 degrees
Angle 2: 50.00 degrees
```